Modeling Personal Security Perception Based on Public Sentiment and Casualty Data

Reut Berkowitz
dept. Biomedical Engineering
Ben-Gurion University of the Negev
Be'er Sheva, Israel
reutfru@post.ac.il

Maayan Kleinman
dept. Biomedical Engineering
Ben-Gurion University of the Negev
Be'er Sheva, Israel
maayankl@post.ac.il

***Abstract*—This study examines whether public perception of personal security during an ongoing conflict can be predicted using a small set of attitudinal and contextual features. We analyzed 14 monthly public opinion surveys conducted between February 2024 and March 2025, focusing on the percentage of respondents reporting high or very high personal security. The predictive variables included five additional sentiment-based survey responses, weekly military fatality counts, and the number of months since the war began. Two regression models were implemented: Ridge Regression and Gaussian Process Regression (GPR). While Ridge Regression outperformed GPR, both models exhibited limited predictive power, with negative R^2 values and low generalization performance. These results highlight the challenges of applying supervised learning with small datasets, but also suggest that simple regularized models may still offer partial insights. Future work with expanded datasets and alternative modeling approaches may yield improved results and deeper understanding of public sentiment dynamics.**

I. Introduction

The ongoing war in Israel has brought about substantial psychological and emotional challenges for the civilian population. Public trust, perceived safety, and emotional resilience have been repeatedly assessed through national surveys [1], yet there remains a need to understand the driving factors behind shifts in these sentiments. In this study, we focus specifically on perceived personal security and examine to what extent it can be predicted based on a combination of contextual and attitudinal variables.

We construct a machine learning framework to investigate whether individual feelings of safety can be inferred from weekly fatality counts, the number of months since the war began, and responses to five additional public opinion questions. These questions address optimism, solidarity, trust in the military, social concern, and expectations regarding the outcome of the war. By treating these five responses as predictive features, alongside time and casualty data, we aim to build regression models that approximate public sentiment.

Our approach integrates data from recurring public surveys conducted by the Institute for National Security Studies (INSS) [1] and official fatality statistics [2]. Unlike prior studies that analyzed public responses descriptively, we apply supervised learning methods to evaluate whether these features can meaningfully predict changes in perceived security. This methodology allows us to explore which types of information are most strongly associated with personal emotional responses during an ongoing national crisis.

II. Methods

This study draws upon 14 monthly public opinion surveys conducted by the Institute for National Security Studies (INSS) between February 2024 and March 2025 [1]. Each survey included a consistent question assessing respondents' perceived personal security, with answers ranging from "Very Low" to "Very High". We consolidated the two most positive responses into a binary indicator representing positive sentiment. The target variable was defined as the proportion of respondents who expressed high or very high levels of personal security in each survey.

To construct the predictive features, we extracted the number of Israel Defense Forces (IDF) casualties reported during the week preceding each survey and computed the number of months elapsed since the outbreak of war in October 2023 [2]. In addition, we incorporated responses to five other survey questions [1] addressing national sentiment: trust in the IDF, belief in victory in Gaza, perceived societal solidarity, social concern, and optimism regarding national recovery. For each question, the two most positive responses were aggregated into a binary indicator, and the percentage of positive responses was used as a numerical feature.

The full feature set thus included seven variables per observation: five public sentiment indicators, the monthly fatality count, and the time since the war's onset. We used these features to predict the percentage of respondents reporting high personal security. The dataset was split using a 70-30 train-test ratio with a fixed random seed for reproducibility. All features were standardized based on the training set.

We implemented two regression models: Ridge Regression and Gaussian Process Regression (GPR). Ridge Regression was selected for its simplicity and robustness under small sample sizes, while GPR was chosen for its ability to capture non-linear relationships and quantify prediction uncertainty. Although we initially experimented with neural networks, they were excluded from the final analysis due to overfitting and instability caused by the limited dataset.

Model performance was evaluated using the coefficient of determination (R^2) and mean squared error (MSE) on the test set. In addition, GPR predictions were plotted with their confidence intervals to illustrate uncertainty. Residual analysis was used to compare error distributions between models and to assess systematic prediction biases.

III. Results

We evaluated the predictive performance of two regression models-Ridge Regression and Gaussian Process Regression (GPR)-in estimating the monthly proportion of respondents who reported a high level of personal security. The feature set included seven variables per month: five survey-based sentiment indicators, the number of military fatalities during the week preceding the survey, and the number of months since the war began.

Ridge Regression achieved an R^2 of -0.225 and a mean squared error (MSE) of 0.000593 on the test set, indicating that its predictions were slightly worse than using the mean value as a fixed predictor. GPR performed even more poorly, with an R^2 of -0.973 and an MSE of 0.000956. Despite their limitations, Ridge Regression consistently produced closer predictions than GPR. Residual analysis revealed that GPR had larger and more erratic deviations, while Ridge errors were smaller and more symmetric.

Figure 1 compares the predicted and true values of perceived personal security across the test samples using both models. The diagonal line represents perfect predictions. As illustrated in the figure, Ridge Regression predictions tend to cluster closer to this ideal line, suggesting lower prediction error. In contrast, GPR predictions deviate further and often underestimate the true values. These patterns are consistent with the performance metrics, confirming that Ridge outperformed GPR in this limited-data setting.

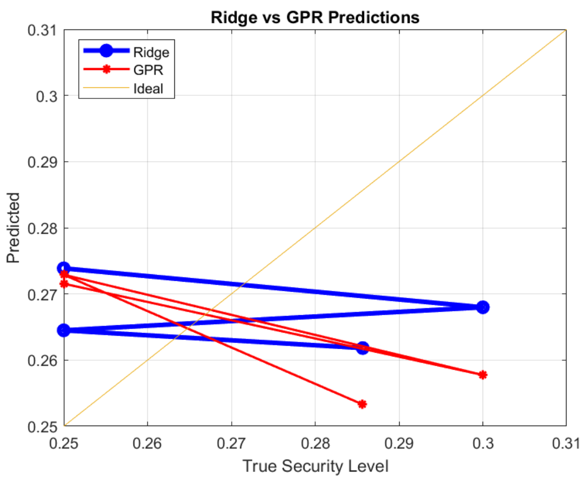

***Figure 1**. Predicted vs. true values of personal security for Ridge and GPR models.*

IV. Discussion, Conclusion and Future Work

In this project, we explored the task of predicting public perception of personal security using a small dataset of monthly surveys conducted during a prolonged conflict. The feature set included five additional public sentiment indicators, along with the number of military fatalities in the previous week and the number of months since the war began. The goal was to estimate the proportion of individuals reporting a high sense of personal security.

We applied two regression models: Ridge Regression and Gaussian Process Regression (GPR). Ridge is a regularized linear model designed to handle multicollinearity and prevent overfitting, making it especially suitable for small datasets with several moderately important features. GPR, on the other hand, is a more flexible non-linear model that estimates uncertainty in predictions. However, its performance was significantly worse in our case, likely due to overfitting caused by the very limited number of samples (n = 14). The Ridge model, while still showing a negative R^2, performed comparatively better and had lower mean squared error (MSE).

The limited data size is the main constraint in this study. With so few samples, even simple models struggle to generalize, and model evaluation is highly sensitive to individual data points. Based on the scikit-learn algorithm selection flowchart, models like Ridge Regression and Support Vector Regression (SVR with linear kernel) are recommended for small regression datasets with several features. We selected Ridge accordingly and avoided ensemble methods, which tend to fail when training data is limited.

In future work, we plan to explore additional regularized models such as ElasticNet, and compare them with SVR using linear or radial kernels. These models, as suggested by standard learning guides, may offer a good balance between complexity and generalization. Moreover, increasing the number of samples-either by collecting more surveys or by switching to weekly rather than monthly data-could open the door for more advanced techniques such as ensemble regression or neural networks, which currently cannot be justified.

To conclude, our findings show that even a small number of contextual and attitudinal variables can provide partial signals about changes in perceived security over time. However, further development requires more data, careful feature selection, and consideration of model simplicity to avoid overfitting in low-sample scenarios.

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RB&MK contributed equally to all parts of this project.

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